

LOAD STABILITY AND CENTER OF GRAVITY TOOLBOX TALK

A load may appear balanced while its center of gravity is somewhere unexpected.

PURPOSE AND SCOPE

Prevent rotation, shifting and instability caused by incorrect center-of-gravity assessment during lifting.

KEY HAZARDS

1  Incorrect center of gravity.	2  Uneven load distribution.	3  Sudden load rotation.	4  Rigging slippage.
5  Movement of internal liquids.	6  Incorrect lifting points.	7  Inadequate trial lift.	8  People beneath the load.

CRITICAL POINTS TO VERIFY IN THE FIELD

01 Incorrect center of gravity.	02 Uneven load distribution.	03 Sudden load rotation.
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REMEMBER — SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

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FIELD REQUIREMENTS AND VERIFICATION

Prevent rotation, shifting and instability caused by incorrect center-of-gravity assessment during lifting.

SERNEM FIELD 01

FIELD VERIFICATION 1

Load weight known

SERNEM FIELD 02

FIELD VERIFICATION 2

Center of gravity assessed

SERNEM FIELD 03

FIELD VERIFICATION 3

Lifting points suitable

SERNEM FIELD 04

FIELD VERIFICATION 4

Rigging correct

VERIFICATION BEFORE START

- | | |
|--|---|
| ☐ Load weight known | ☐ Center of gravity assessed |
| ☐ Lifting points suitable | ☐ Rigging correct |
| ☐ Trial lift planned | ☐ Load behaviour monitored |

SUPERVISOR DISCUSSION POINTS

- Where is the center of gravity?
- Are lifting points suitable?
- Is the load asymmetrical?

FIELD APPLICATION



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CRITICAL CONTROL MEASURES

1. ELIMINATE / REDUCE THE RISK

Determine load weight and center of gravity. • Observe load behaviour during the trial lift. • Stop lifting if the load is unstable.

2. ENGINEERING & PHYSICAL CONTROLS

Use approved lifting points. • Apply additional rigging control for asymmetrical loads. • Keep all personnel clear of suspended loads.

3. WORK METHOD & FIELD CONTROLS

Configure rigging around the center of gravity. • Assess movement of liquids inside equipment.

4. PPE, VERIFICATION & SUPERVISION

Perform a low-height trial lift. • Plan safe tag-line use where required.

CRITICAL WARNING



Team, we will lift according to both load weight and center of gravity. If the load is unstable during the trial lift, we will stop and re-rig.

SITE CHECKLIST — BEFORE STARTING WORK

- | | |
|--|---|
| ☐ Load weight known | ☐ Center of gravity assessed |
| ☐ Lifting points suitable | ☐ Rigging correct |
| ☐ Trial lift planned | ☐ Load behaviour monitored |
| ☐ Asymmetry assessed | ☐ Liquid movement assessed |
| ☐ Area below clear | ☐ Crew briefed |



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ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
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2			
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